

Chapter 1

Homework Hints

Chapter ??

- There was a typo in problem 3.2: the problem should have read “...but $x^3 + x - 4 \notin C$ ” not “...but $x^3 + x - \notin C$ ”.

Definition 1. **[Exercise ??]** Let $\mathbb{R}[x]$, the set of all polynomials with real coefficients. Let C be the subset of $\mathbb{R}[x]$ with constant terms that are 0. For example, $x^3 + x \in C$ but $x^3 + x - 4 \notin C$. Prove that C is or is not an ideal.

Hint: You should multiply a few polynomials and see what happens. If it is an ideal, it should pass the ideal test (theorem ??). If it is not, a simple example will suffice. Theorem ?? is an “if and only if” statement. So if part 1 or 2 is not met, then the non-empty subset is not an ideal. Keep in mind that for part 2, the $r \in R$ means any element in the ring. For this problem, that would include polynomial with non-zero constants like $x^3 + x - 4$, $x - 2$, 3, etc.

Definition 2. **[Exercise ??]** Let R be a ring with unity 1_R and let I be an ideal of R . If I has an element x and x^{-1} such that $x \cdot x^{-1} = x^{-1} \cdot x = 1_R$ ¹, prove that $I = R$.

Hint: We already know that $I \subseteq R$ so we only need to show that $R \subseteq I$. Key here is that 1_R is in I , and that $r = 1 \cdot r$.

¹Such an element x is called a *unit*.